3500 Engine Exhaust Manifold Salvage

Maintenance and Repair | Application | Component Life Management | **Component Renewal (CRC)** | MARC Management

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1.0 Introduction

This document contains information on how to salvage the exhaust manifold of 3500 engines by using metal spray technology.

Figure 1. Typical 3500 Exhaust Manifold

2.0 Best Practice Description

This new process has been developed at MATCO machine shop to salvage the 3500 exhaust manifold ends when fretting and pitting exists. The salvage procedure is an alternative that is more durable than a machined sleeve.

The salvage procedure consists of pre-machining the exhaust manifold to remove the damage, thread cutting the surface to be salvaged, metal spraying the surface, and finally machining the outside diameter.

3.0 Implementation Steps

3.1. Pre-Machining: The exhaust manifolds must be pre-machined to remove the pitting on it, most of the time cutting off 0.040” to the outside diameter is adequate. After pre-machining is complete, machine a 20 threads per inch pattern on the surface to create the bond interface for the spray. The total depth of thread cut should be around 0.020” - 0.030” and if thread cut done on the workpiece, consider the usage of grit blasting not mandatory.

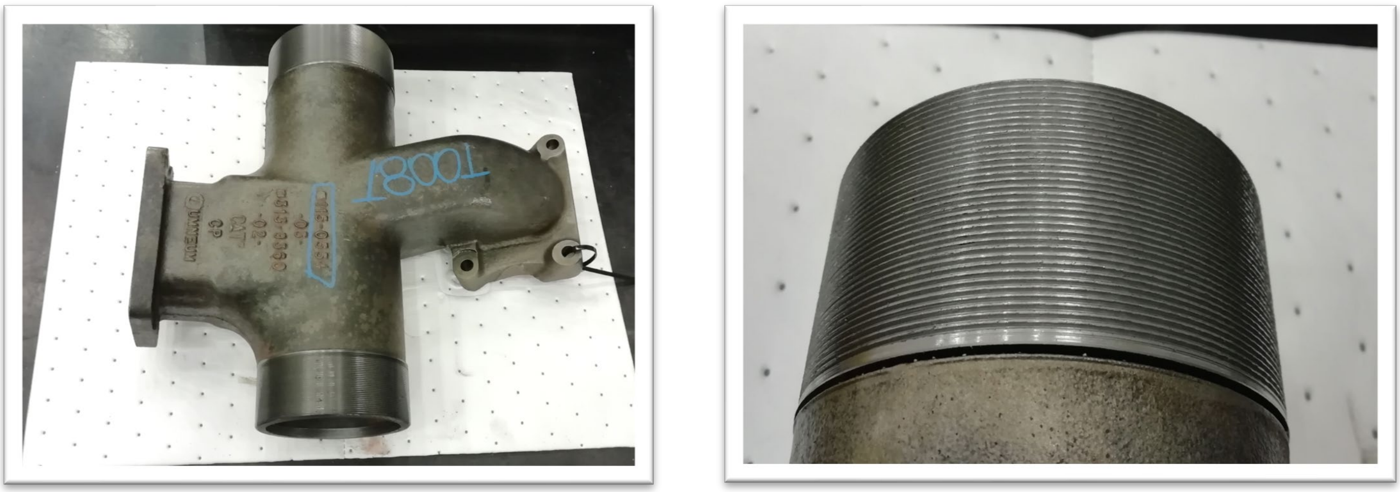

Figure 2. Exhaust Manifold (Thread Cutting)

3.2. Surface protection: Use anti-bond paint to protect the areas you do not want to be sprayed.

Figure 3. Exhaust Manifold (Surface Protection)

3.3. Metal Spraying: Can be performed by either an industrial robot or by hand. Consumable wire used Oerlikon Metco 8400 or equivalent. For spraying parameters please see product datasheet.

Figure 4. Exhaust Manifold (Sprayed)

3.4 Finishing cut: Use a horizontal lathe to machine workpiece to desired specifications.

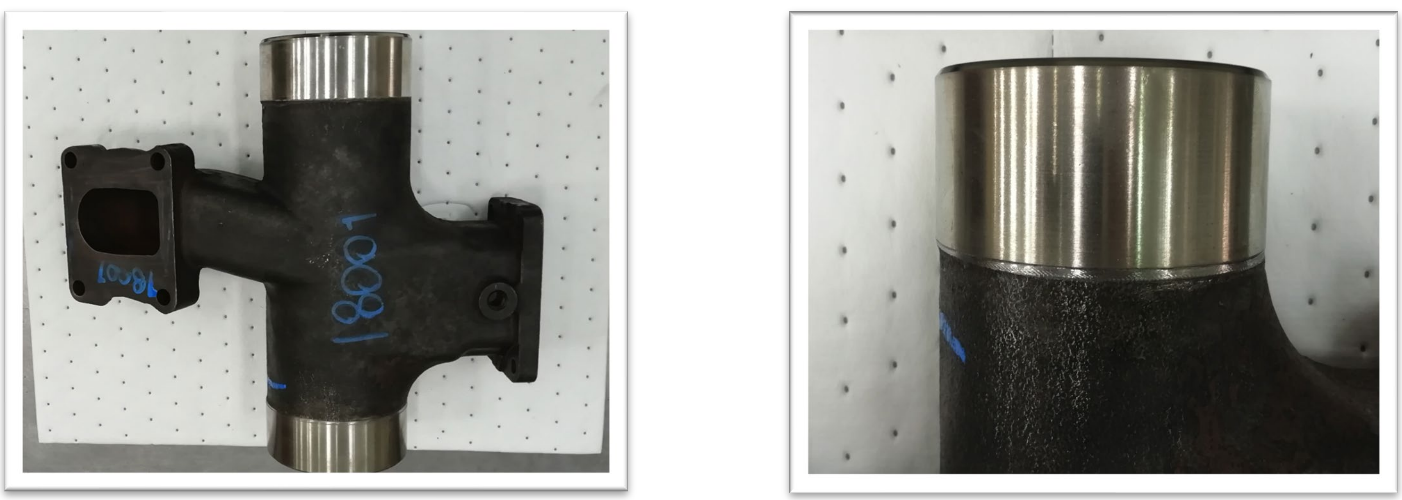

Figure 5. Exhaust Manifold (Machined)

4.0 Benefits

4.1. Increases the lifetime of the salvage using wear resistant materials to create the coating.

4.2. Eliminates the need to install a machined sleeve when pitting exists on exhaust manifolds

4.3. Reduces labor time.

5.0 Resources Required

5.1. Conventional or CNC horizontal lathe

5.2 Industrial robot (optional).

5.3 Metal spray equipment.

6.0 Supporting Attachments / References

SEBF2125 – Thermal Spray Procedures for Exhaust Manifold (I.D. & O.D.)

7.0 Related Best Practices

None

8.0 Acknowledgements

This Best Practice was written by:

Name: Oscar De La O / Machine Shop Engineer
E-mail: odelao@matco.com.mx
Company: Empresas MATCO S.A. de C.V.

Recognition:

Team:
Lauro Alvarado / Machine Shop Chief
Julián Pintado / Machine Shop Engineer
Oscar De La O / Machine Shop Engineer